Code Sense

Deployment & Security Dossier

Offline AI Code-Review & SBOM Scanner (Desktop)

Document version: v1.5
Date: 2026-05-28
Classification: Confidential
Prepared for: Bank Information Security / Vendor Risk / Endpoint Engineering
Deployment model: Single self-contained Windows desktop application, fully offline

*This document contains the product Bill of Materials, security posture, data-handling model, and risk register for review prior to endpoint deployment. Items marked **[VERIFY]** require vendor confirmation before sign-off.*

1. Executive summary

Code Sense is a security code-review tool that runs **entirely on the user's laptop with no network connectivity**. It scans source code (SAST, via a bundled local AI model) and software dependencies (SCA/SBOM, via bundled Syft/Grype) and stores all results in a local database. **No source code, scan results, or telemetry ever leaves the machine** — there is no cloud backend, no external AI/inference API, and no licensing "phone home."

This is the primary control relevant to a bank: code under review (potentially proprietary or regulated) is processed **locally only**. The application binds exclusively to the loopback interface (`127.0.0.1`) and makes **zero outbound or inbound network connections** at runtime.

The package is delivered as a signed Windows installer that bundles all runtime components (application, AI model, scanner binaries, embedded database, web runtime). It is licensed for a fixed **90-day** term from first launch, after which it enters a read-only state.

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2. Product overview

Attribute	Detail
Function	SAST (AI code review) + SCA/SBOM (dependency CVE + license scanning)
Form factor	Native Windows desktop app (Tauri shell hosting a local web UI)
Backend	Django (Python) served by waitress on `127.0.0.1:8585` (loopback only)
AI inference	Local llama.cpp server (`127.0.0.1`, loopback only) — no external API
Datastore	Embedded SQLite file in the per-user profile
External services	**None** at runtime (no cloud, no MongoDB, no central license server)
Multi-user	Single-user desktop; in-app RBAC (admin/manager/user)

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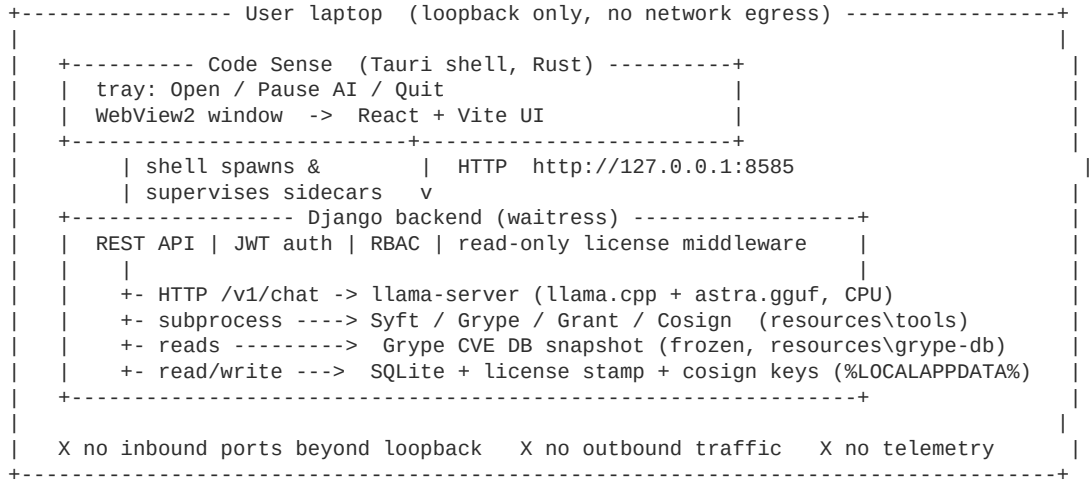
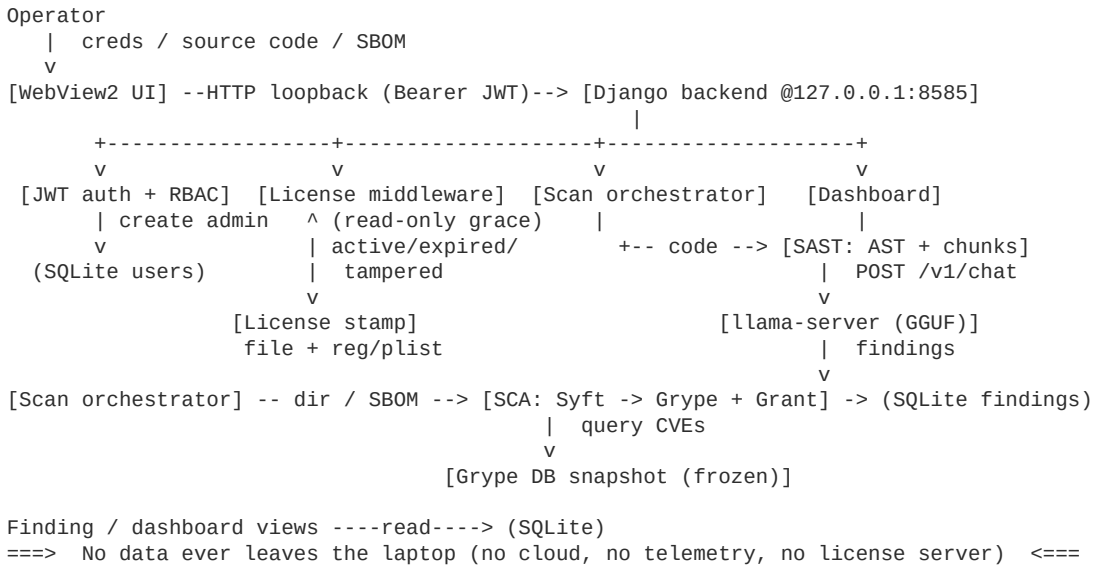
3. Deployment architecture & data flow

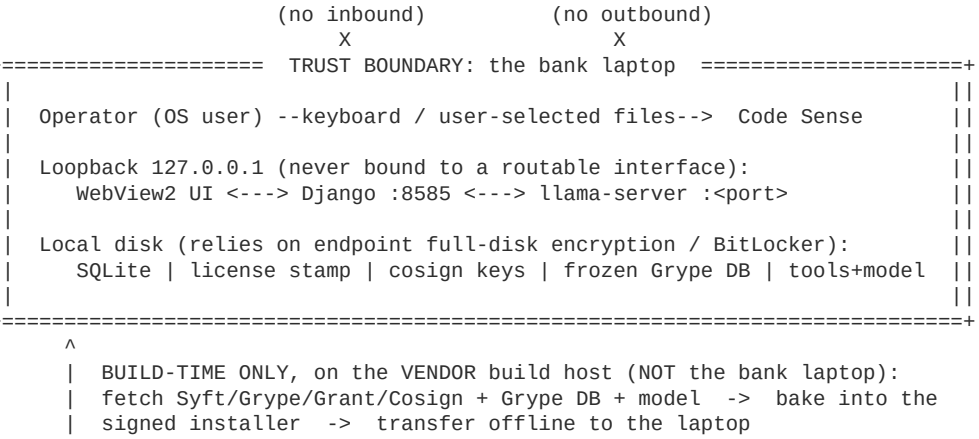
3.1 Architecture (component / deployment)



3.2 Data flow (all local)



3.3 Network / trust boundary



Data flow (all local):

1. User selects source (ZIP upload or local path) or an SBOM file inside the app.
2. Backend runs AST metrics + the local AI model (SAST) and/or Syft->Grype (SCA) — all on-device.
3. Findings (CVEs, CWEs, licenses, severities) are written to the local SQLite database.
4. The UI reads them back over loopback. Reports can be exported to local files (Excel/CSV) by the user.

No data egress path exists. The application contains no code that opens outbound sockets to non-loopback hosts at runtime. (Network access occurs only on the **vendor's build machine** when assembling the installer — see §7.9 — never on the bank laptop.)

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4. Network & connectivity requirements

Direction	Requirement
Outbound (internet)	**None.** Works fully air-gapped.
Outbound (LAN)	**None.**
Inbound	**None.** Listeners bind to `127.0.0.1` only (`ALLOWED_HOSTS=127.0.0.1,localhost`).
Loopback (same host)	`127.0.0.1:8585` (backend), one private loopback port for the AI server.
Proxy/firewall changes	**Not required.** No allow-listing needed.

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5. Data handling, privacy & residency

Topic	Detail
Data processed	Source code / repositories and SBOM files the user submits; scan findings.
Data residency	100% local to the laptop. Nothing is transmitted off-device.
Storage location	`%LOCALAPPDATA%\com.codesense.desktop\` (SQLite DB, license stamp, signing keys, logs).
PII	Only the operator account (name, email, bcrypt-hashed password) entered during setup. No end-customer PII is collected by the tool.
Telemetry / analytics	**None.** No usage data, crash reporting, or beaconing.
Third-party data sharing	**None.**
Retention / deletion	Data persists in the local SQLite file until the user deletes records or uninstalls; uninstall + removal of the data directory fully purges it.
Encryption at rest	Relies on the endpoint's full-disk encryption (e.g., BitLocker) per bank policy. The app does not implement its own DB encryption. **[VERIFY]** confirm BitLocker is enforced on target laptops.

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6. Authentication, authorization & application security

Control	Implementation
Authentication	JWT (HS256, 60-min expiry); credentials verified with **bcrypt**.
First-run setup	On first launch the user creates the admin account in-app. **No default credentials are shipped.** A strong-password policy is enforced (min 8 chars; upper/lower/digit/special).
Authorization	Role-based access control (admin / manager / user) with per-action permission flags enforced server-side via decorators.
Session secret	Django `SECRET_KEY` is **generated uniquely per install** on first run and persisted locally (no shared/hardcoded key).
Transport	Loopback only; traffic does not traverse any network.
Hardening	`DEBUG=False` in the shipped build; `ALLOWED_HOSTS` restricted to loopback; CORS restricted to the app's own origin.
Independent review	An internal security review of the codebase found **no exploitable vulnerabilities** (no SQL/command injection, no unsafe deserialization, parameterized ORM throughout). Defense-in-depth hardening was applied to reviewed items (segment-aware request gating, per-install license key derivation, atomic admin bootstrap). See §11.

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7. Bill of Materials (BOM / SBOM)

A machine-readable **CycloneDX SBOM can be generated on demand** using the Syft binary bundled with the product (the same tool the app uses for scanning). The tables below enumerate the first-party components and all bundled third parties with versions, licenses, and sources.

7.1 First-party application components

Component	Description	Language
Code Sense backend	REST API, scan orchestration, RBAC, licensing	Python / Django
Code Sense frontend	Local web UI	TypeScript / React
Tauri desktop shell	Window, tray, sidecar process management	Rust
AI code-review model	Fine-tuned LLM ("astra-code-reviewer")	model weights (GGUF)

7.2 Backend runtime dependencies (Python 3.13)

All pinned to the exact versions resolved into the shipped build. All OSI-approved / permissive.

Package	Version	License
Django	5.2.14	BSD-3-Clause
djangorestframework	3.15.2	BSD-3-Clause
django-cors-headers	4.9.0	MIT
PyJWT	2.13.0	MIT
bcrypt	4.3.0	Apache-2.0
cryptography	44.0.3	Apache-2.0 / BSD-3-Clause
waitress	3.0.2	ZPL-2.1
requests	2.34.2	Apache-2.0
httpx	0.27.2	BSD-3-Clause
openpyxl	3.1.5	MIT
python-dotenv	1.2.2	BSD-3-Clause
asgiref	3.11.1	BSD-3-Clause
sqlparse	0.5.5	BSD-3-Clause
urllib3	2.7.0	MIT
certifi	2026.5.20	MPL-2.0
idna	3.16	BSD-3-Clause
charset-normalizer	3.4.7	MIT
anyio	4.13.0	MIT
sniffio	1.3.1	MIT / Apache-2.0
h11	0.16.0	MIT
httpcore	1.0.9	BSD-3-Clause
cffi	2.0.0	MIT
pycparser	3.0	BSD-3-Clause
dnspython	2.8.0	ISC
et-xmlfile	2.0.0	MIT

Note: `pymongo`/`bson` were **removed** during the migration to a self-contained SQLite store;

no MongoDB driver or server is present. (`certifi`/MPL and `waitress`/ZPL are used unmodified as libraries; no copyleft obligation is triggered for this usage.)

7.3 Frontend dependencies (Node / bundled into static assets; key direct deps)

Package	Version	License
react / react-dom	19.1.0	MIT
@tanstack/react-router	1.123.2	MIT
@tanstack/react-query	5.83.0	MIT
@tanstack/react-form	1.14.1	MIT
axios	1.10.0	MIT
@radix-ui/react-* (dialog, select, dropdown, tooltip, switch, collapsible, slot)	1.x–2.x	MIT
tailwindcss	4.1.11	MIT
recharts	3.0.2	MIT
lucide-react	0.525.0	ISC
sonner	2.0.5	MIT
clsx / class-variance-authority / tailwind-merge / next-themes	current	MIT
vite (build)	7.0.0	MIT
typescript (build)	5.8.3	Apache-2.0

The full transitive dependency tree is pinned by `client/package-lock.json` / `bun.lock` and is available as a CycloneDX SBOM on request.

7.4 Desktop shell (Rust / Tauri crates)

Crate	Version	License
tauri	2.x	MIT / Apache-2.0
tauri-plugin-shell	2.x	MIT / Apache-2.0
tauri-build	2.x	MIT / Apache-2.0
serde / serde_json	1.x	MIT / Apache-2.0

Full crate lock is pinned by `client/src-tauri/Cargo.lock` (generated at build).

7.5 Bundled third-party executables

Binary	Version	Purpose	License	Source
llama-server (llama.cpp)	release `[VERIFY: record build tag b####]`	Local AI inference engine	MIT	github.com/ggml-org/llama.cpp
Syft	1.18.1	SBOM generation	Apache-2.0	github.com/anchore/syft
Grype	0.85.0	Dependency CVE scanning	Apache-2.0	github.com/anchore/grype
Grant	0.2.4	License compliance scanning	Apache-2.0	github.com/anchore/grant

Binary	Version	Purpose	License	Source
Cosign	2.4.1	SBOM signing/verification	Apache-2.0	github.com/sigstore/cosign
WebView2 Runtime (fixed version)	per Microsoft	Web rendering host	Microsoft Software License (redistributable)	Microsoft

7.6 AI model

Attribute	Detail
Name	"astra-code-reviewer" (fine-tune)
Base architecture	Qwen2.5-Coder-3B (`Qwen2ForCausalLM`): hidden 2048, 36 layers, 16 heads, vocab 151,936, context 32,768
Shipped format	GGUF, **q8_0** quantization
File	`astra.gguf` (bundled), 3.1 GB
SHA-256	`e927148c7023a5f5c67ad508cbbb3613a5a9a5cbac4b09a0a999a1b8f59b0586`
Function	Generates structured vulnerability findings from code chunks; runs 100% locally
[VERIFY] Licensing	The base **Qwen2.5-Coder-3B** is published under the **Qwen license** (the 3B variant historically uses a Qwen Research/Community license, not Apache-2.0). The vendor must confirm commercial-use rights for bank deployment and provide the model license text.

7.7 Embedded datastore

Component	Version	License
SQLite (via Python `sqlite3`)	bundled with Python 3.13	Public Domain

7.8 Artifact integrity

Artifact	Integrity control
AI model `astra.gguf`	SHA-256 published above (verify post-transfer).
Generated SBOMs	Signed with Cosign (Ed25519) at scan time; verifiable offline.
Windows installer	**[VERIFY]** Authenticode code-signed by the vendor's certificate (provide cert details + thumbprint).
Final installer SHA-256	**[VERIFY]** produced on the Windows build host; record and ship alongside the installer.

7.9 Build-time provenance (vendor side — not on the bank laptop)

The installer is assembled on the vendor's build host: dependencies are pinned, the AI model is quantized from the base weights, the scanner binaries + Grype vulnerability-DB snapshot are downloaded, and everything is bundled and signed. **None of these network operations occur on the bank endpoint** — the bank receives only the finished, signed installer.

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8. System requirements & installation

Item	Requirement
OS	Windows 10/11 (x64)
CPU	x64; AI scans are CPU-only (no GPU required). More cores = faster scans.
RAM	8 GB minimum, 16 GB recommended (the AI model uses ~3–4 GB while active).
Disk	~4–5 GB installed (model + binaries + runtime).
Privileges	Per-user install (no kernel drivers, no services). Admin rights not required to run. **[VERIFY]** confirm whether your packaging policy needs a per-machine vs per-user MSI/NSIS.
Runtime prerequisites	Bundled (Python runtime, WebView2 fixed runtime). No separate installs.
Install footprint	App in Program Files (or per-user); data in `%LOCALAPPDATA%\com.codesense.desktop\`.

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9. Licensing & entitlement

Topic	Detail
Term	**90 days from first launch** (configurable at build).
Mechanism	Offline. A signed "first-seen" timestamp is recorded locally; no server contact.
Expiry behavior	**Read-only grace** — after expiry, existing findings remain viewable but new scans and edits are disabled.
Tamper resistance	A signed local stamp (file + registry) plus a system-clock-rollback guard. **Honest scope:** this is deterrence against casual date-changing/copying, **not** cryptographic DRM against a determined attacker.
Renewal	A new build/license is issued by the vendor (offline); no network dependency.

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10. Operations & lifecycle

Topic	Detail
Start / stop	Launched from Start Menu; a system-tray menu offers **Open / Pause AI engine (frees RAM) / Quit**. Closing the window minimizes to tray; Quit cleanly stops all components.
Logs	Written locally under the data directory; no remote log shipping.
Backup	Back up `%LOCALAPPDATA%\com.codesense.desktop\app.sqlite3` to retain findings.
Updates	Delivered as a new signed installer (no auto-update / no network update channel).
Uninstall	Standard Windows uninstall; delete the data directory to purge all local data.

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11. Internal security assessment summary

A focused security review of the application changes was performed (injection, authn/authz, crypto/secrets, deserialization, data exposure).

- **Result:** no concretely-exploitable vulnerabilities identified.
- Database access uses the parameterized Django ORM (no raw SQL); subprocess calls to scanner binaries use argument lists (no shell interpretation); no `pickle`/`yaml`/`eval` deserialization;

the React UI uses no unsafe HTML sinks.

- **Hardening applied:** request gating uses exact path-segment matching; the offline-license key is derived per-install from the unique `SECRET_KEY` rather than a shared constant; first-admin creation enforces password strength and is transaction-guarded.
- **Test coverage:** 59 automated backend tests pass.

Vulnerability management note: Grype CVE data is bundled as a **point-in-time snapshot** taken at build. It does not auto-update offline; refreshed CVE data requires a new build. (This is a deliberate trade-off for the air-gapped model.)

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12. Risk register & limitations (full disclosure)

#	Item	Impact	Mitigation / status
1	**Model license [VERIFY]**	Legal/compliance	Vendor to confirm Qwen2.5-Coder-3B commercial-use rights + supply license text before sign-off.
2	Installer code-signing **[VERIFY]**	Endpoint trust	Vendor to Authenticode-sign and provide certificate details + installer SHA-256.
3	License tamper model is deterrence, not DRM	Entitlement	Accepted by design; not a data-security risk.
4	Grype CVE DB is frozen at build	Stale CVE coverage	Refresh via rebuild on a defined cadence (e.g., quarterly).
5	AI model is **q8_0** (~3.1 GB)	Larger/slower than q4	Functional; a smaller Q4_K_M build is available if size/perf matters.
6	DB encryption relies on endpoint FDE	Data at rest	Confirm BitLocker policy on target laptops.
7	Windows installer build not yet executed/validated in this package	Delivery readiness	Build + smoke-test on a Windows host per the included `BUILD.md` before distribution.
8	AI findings may contain false positives/negatives	Review quality	Tool augments, does not replace, human review; findings are advisory.

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13. Compliance mapping (summary)

Control area	How Code Sense addresses it
Data residency / sovereignty	All processing and storage on-device; no egress.
Third-party data sharing	None.
Network exposure	No listening ports beyond loopback; no outbound traffic.
Access control	Per-install admin bootstrap, RBAC, bcrypt, JWT.

Control area	How Code Sense addresses it
Supply-chain integrity	Pinned dependencies; bundled SBOM tooling; Cosign-signed SBOMs; (vendor) Authenticode-signed installer.
Auditability	Local findings DB + logs; CycloneDX SBOM available on request.
Least privilege	Per-user, no drivers/services, admin not required to run.

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14. Support & contacts

Item	Detail
Vendor	Astra (repository: github.com/grv-astra/yacm) **[VERIFY contact/SLA]**
Support channel	**[VERIFY]**
Patch/update cadence	**[VERIFY]** (recommend quarterly rebuild for refreshed CVE data)
Escalation	**[VERIFY]**

Appendix A — Regenerating a machine-readable SBOM

The product bundles Syft; an authoritative CycloneDX SBOM of any component can be produced offline:

```
syft dir:<path-to-component> -o cyclonedx-json=code-sense-sbom.json
cosign sign-blob code-sense-sbom.json --key cosign.key --bundle code-sense-sbom.bundle.json
```

Appendix B — Exact backend dependency manifest

The Python manifest in §7.2 is the live `pip freeze` of the shipped virtual environment (Python 3.13.1). The frontend (`package-lock.json`/`bun.lock`) and Rust (`Cargo.lock`) lockfiles pin the complete transitive trees and can be provided in full on request.